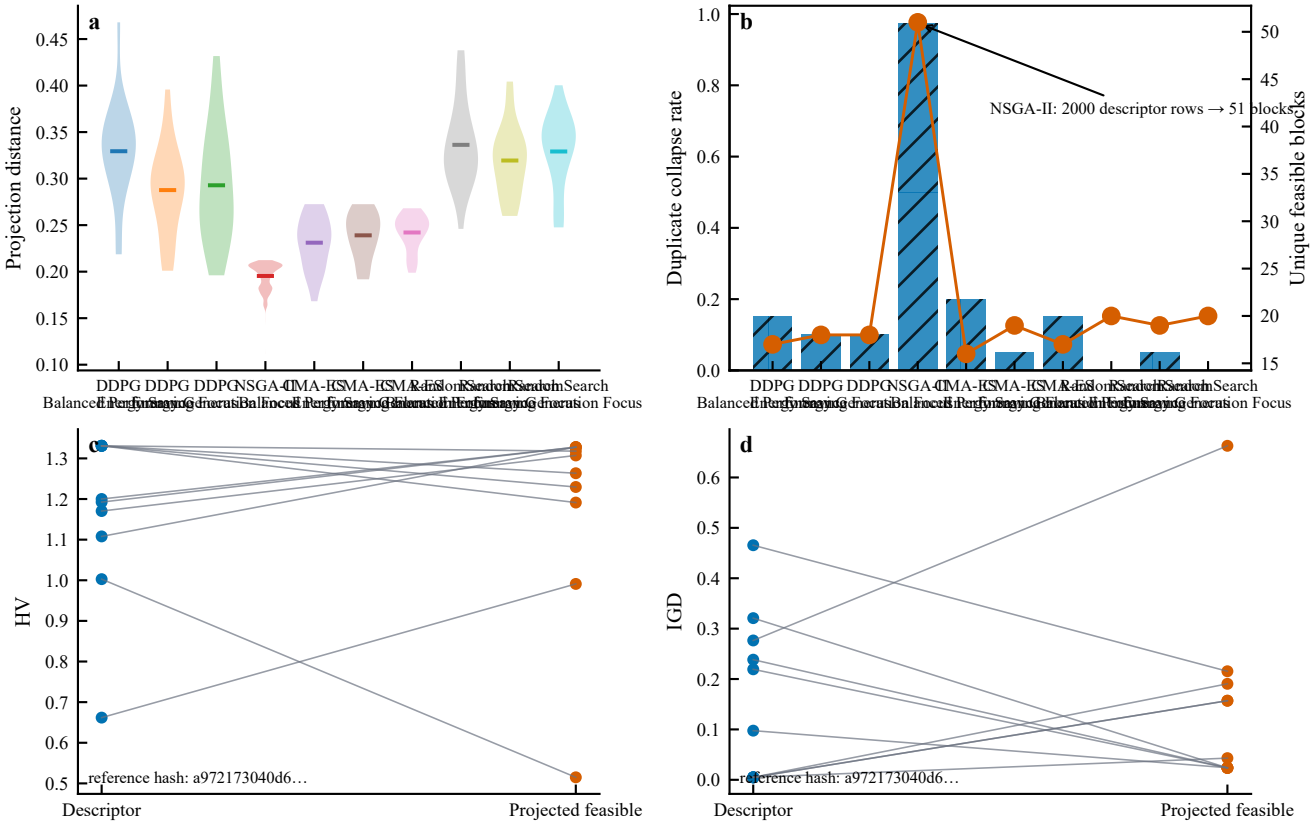
a
Projection distance
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
b
Duplicate collapse rate
Unique feasible blocks
NSGA-II: 2000 descriptor rows → 51 blocks
1.0
0.8
0.6
0.4
0.2
0.0
50
45
40
35
30
25
20
15
DDPG Balanced
DDPG Energy Focus
DDPG Sensing Focus
NSGA-II
CMA-ES Balanced
CMA-ES Energy Focus
CMA-ES Sensing Focus
Random Search Balanced
Random Search Energy Focus
Random Search Sensing Focus
c
HV
1.3
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
reference hash: a972173040d6…
Descriptor
Projected feasible
d
IGD
0.6
0.5
0.4
0.3
0.2
0.1
0.0
reference hash: a972173040d6…
Descriptor
Projected feasible